

# Anderson Steward González Rivera

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I am a hybrid biotechnologist with a strong background in molecular biology and bioinformatics. I have gained international research experience in Sweden and Mexico and successfully led two teams in the iGEM Design League. Additionally, I have presented my academic work at several international scientific events. My professional experience includes molecular diagnostics for infectious diseases and hands-on involvement in omics approaches across various research projects. I am highly motivated to integrate my knowledge and skills to address challenges in bioinformatics and biotechnology.

## Education

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- **Bachelor's Degree in Biotechnology**  
*University Institution Colegio Mayor de Antioquia, Faculty of Health Sciences. Final grade of 4.3/5.0, Recognised by the Ministry of Education for one of the highest national scores in the Saber Pro state examination, Medellín, Colombia.* 12/2023
- **Academic High School Diploma**  
*José María Bernal High School. Medellín, Colombia.* 11/2018

## Internships and Professional Experience

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- **Laboratory Assistant Molecular Biology.**  
*On-Site Diagnostic OSD.*  
*Process and analyses biological samples for molecular diagnostics (3000 samples to date). Activities include the pre-analytical phase of sample preparation, DNA extraction from clinical samples using both manual and automated methods, preparation of reagents for qPCR, and execution of the test using hydrolysis probes (TaqMan). Additionally, analysis, interpretation, and validation of results are performed.* 11/2025 - Currently
- **Freelancer Virtual Teacher.**  
*Superprof.co.*  
*Academic tutor in Molecular Biology, Genetics and Bioinformatics. Provided personalised mentorship in advanced concepts of molecular biology and metagenomics for university-level students.* <https://acortar.link/X6vzCh> 02/2025 – Currently
- **Laboratory Technician**  
*University Institution Colegio Mayor de Antioquia Medellín, Colombia.*  
*Managed biotechnology laboratory operations with over 60 people, including students, researchers, professors, and staff. Also, responsible for delivering laboratory equipment, managing reagents, and preparing non-hazardous solutions.* 10/2024 – 12/2024
- **Young Researcher**  
*University Institution Colegio Mayor de Antioquia, Medellín, Colombia.*  
*Finalized three projects. Some activities required were to standardize DNA extraction, PCR amplification and send the samples for sequencing. Perform bioinformatic, phylogenetic and statistical analysis using R. In addition, my work also included writing the manuscript draft for each project, planning and executing all activities autonomously and leading the study sessions of a study group at the Health Sciences Faculty.* 02/2024 – 12/2024

- Project student (Professional Intern)**  
*Swedish University of Agricultural Sciences, Molecular Virology Group, Uppsala, Sweden.*  
 Executed experiments for the Undergraduate Vacation Bursary of the British Society of Plant Pathology. Activities included statistical analysis, RT-PCR, PCR, DNA/RNA extraction, fluorescence microscopy, molecular cloning, enzyme restriction, electrophoresis, viral transfection and Western blot. A short report, a complete record of my work with data, a short video blog and a presentation at a symposium were needed.
 06/2023 – 11/2023
- Professional Intern**  
*University Institution Colegio Mayor de Antioquia, Medellín, Colombia.*  
 Performed genomic analysis for the project. Some of the activities were creating a Next Generation Sequencing pipeline from the raw sequencing data to the processed data, performing a Genome-Wide Association Study (GWAS) using a Linux environment, plotting and analysing the results using RStudio, and finally modelling and evaluating the protein output using ab initio approaches.
 12/2022 – 06/2023
- Delfín Summer Intern.**  
*IPICYT, Group of Science and Computational Biology, San Luis de Potosí, México.*  
 Conducted bioinformatics analysis. Some of the activities included downloading transcriptomics data from public databases, using RStudio to explore data quality, and performing machine learning predictions.
 06/2022 – 08/2022

#### Courses and Conference Participation

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|---------------------------------------------------------------------------------------------------------------------------|---------|
| • Artificial Intelligence-Advanced Deep Learning                                                                          | 03/2026 |
| • Cell culture course                                                                                                     | 10/2025 |
| • Fundamentals of generative artificial intelligence                                                                      | 09/2025 |
| • Advanced Python.                                                                                                        | 08/2025 |
| • Essential Python                                                                                                        | 07/2025 |
| • Academic Day: Bioinformatics Applied to the Molecular Epidemiology of Infectious Pathogens.                             | 05/2025 |
| • Second International Symposium on Clinical Microbiology and Infectious Diseases.                                        | 12/2024 |
| • University Diploma in University Teaching.                                                                              | 10/2024 |
| • Presentation at Red Colombiana de Semilleros de Investigación (RedCOLSI) National phase.                                | 10/2023 |
| • Presentation at BSSP Summer Internship Symposium.                                                                       | 09/2023 |
| • Presentation at XVII European Congress of Ichthyology. Czech University of Life Sciences, Prague, Czech Republic.       | 09/2023 |
| • Course of Introduction to Experimental Design and Applied Statistical Analysis.                                         | 08/2022 |
| • Presentation at the International Congress of the XXVII Summer of Scientific and Technological Research of the Pacific. | 07/2022 |
| • Presentation at Red Colombiana de Semilleros de Investigación (RedCOLSI) departmental phase.                            | 05/2022 |
| • Chemoinformatics applied to drug design and discovery                                                                   | 04/2022 |
| • Course of Fundamentals of Genomics for Health Research.                                                                 | 10/2021 |

#### Awards and Achievements

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|------------------------------------------------------------------------------------|---------|
| • Best national Scores at the Examination of Quality of Higher Education Saber Pro | 12/2024 |
| • BSPP Undergraduate Vacation Bursary Award.                                       | 06/2023 |
| • Participation guarantee on the international phase of RedCOLSI.                  | 05/2023 |

- Silver medal iGEM Design League. 11/2022
- Recognition for outstanding participation as a speaker at the national and international phase of RedCOLSI. 10/2022
- Recognition for outstanding participation in the internship, and as a speaker at the International Congress of the XXVII Summer of Scientific and Technological Research of the Pacific. 09/2022
- Participation guarantee on the international phase of RedLASIRC. 07/2022
- Summer research internship DELFIN Award. 06/2022
- Bronze medal iGEM Design League. 11/2021
- First place at the Biotechnology Fair. University Institution Colegio Mayor de Antioquia. Medellín, Colombia. 10/2021
- Published Book chapter ISBN: 978-958-49-2798-9. 05/2021

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### Skills and Hobbies

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| <b>Soft skills:</b>   | Team player, responsible, assertive, resilient, adaptable, disciplined, autonomous, creative, public speaking, presentations, time management, collaborative person, research, critical thinking, detail-oriented, teaching, and curious.                                                                                                 |
| <b>Laboratory:</b>    | Antimicrobial susceptibility testing, sequencing, DNA/RNA extraction and quantification, RT-qPCR, q-PCR, PCR, electrophoresis, molecular cloning, enzyme restriction, bacterial transformation, cell culture, fluorescence microscopy, HPLC.                                                                                              |
| <b>Computational:</b> | Public databases, Linux environment, R programming, Python, statistical analysis, R studio, Benchling, MEGA, Geneious Prime, UGENE, SnapGene, SPSS, BEAST, BEAUi, GROMACS, machine learning, genomics (GWAS, variant calling), transcriptomics (differential expression), proteomics (protein modelling), phylogenetics and metagenomics. |
| <b>Languages:</b>     | Spanish (native), English (B2 Certificate), French (B1 level) and German (A1 level, currently learning).                                                                                                                                                                                                                                  |

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### References

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| <b>Ph.D. Eugene Savenkov</b> | Professor of Immunology, Genetic Engineering, Genetics, and Virology at the Swedish University of Agricultural Sciences.<br>+46 70 386 32 72 - <a href="mailto:eugene.savenkov@slu.se">eugene.savenkov@slu.se</a>                       |
| <b>Ph.D. José Martínez</b>   | Professor of Bioinformatics, Molecular Biology and Experimental Design at the University Institution Colegio Mayor de Antioquia.<br>+57 318 459 4313 - <a href="mailto:jose.martinez@colmayor.edu.co">jose.martinez@colmayor.edu.co</a> |
| <b>Ph.D. Cesar Muñoz</b>     | Scientific director at One Site Diagnostic S.A.S.<br>+57 300 5143626 - <a href="mailto:direccioncientifica@osdiagnostic.com">direccioncientifica@osdiagnostic.com</a>                                                                   |